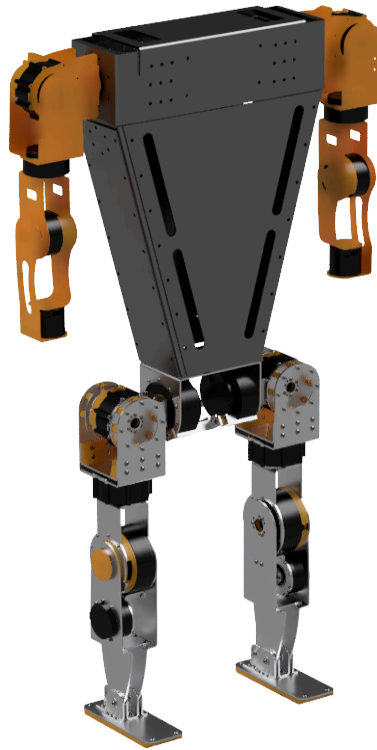


Tritonoid S1



Overview

- ★ Height : 1.3 m
- ★ Mass : 31 kg
- ★ DoF : 20
- ★ Actuation : Robstride quasi-direct-drive
- ★ Compute : Jetson Orin Nano + ESP32



Motors

Model	Peak Torque	Weight	Size	Position	Qty	Unit Price
RS00	14 N·m	0.310 ± 10 kg	57 x 57 x 51 mm	Wrists	2	\$125
RS02	17 N·m	0.405 ± 5 kg	78.5 x 78.5 x 45.5 mm	Foot Elbow Shoulder Yaw	6	\$145
RS03	60 N·m	0.880 ± 20 kg	106 x 106 x 56 mm	Hip Yaw Hip Roll Shoulder Roll Shoulder Pitch	8	\$225
RS04	120 N·m	1.420 ± 20 kg	120 x 120 x 56 mm	Knee Hip Pitch	4	\$255
Total					20	\$3,940

Torso Electronics

Component	Qty	Envelope (mm)	Mounting / Access Notes
Jetson Orin Nano dev kit	1	100 × 79 × 45	Includes fan clearance above. USB, RJ45, HDMI, and DC barrel must be accessible for programming. 4× M2.5 mounting holes.
PDB (custom 4 layer PCB)	1	150 × 100 × 30	Includes buck converter modules (2×), bulk caps, and headers. Needs airflow. XT30 fanout along one edge, leave ~50 mm cable service loop.
ESP32-S3 dev boards	4	55 × 28 × 15 each	Socketed on PDB (preferred) or on a small daughterboard. USB-C

Component	Qty	Envelope (mm)	Mounting / Access Notes
			on each must remain accessible for reflashing without disassembly.
CAN transceiver ICs	4	on PDB	Surface mount on PDB, no separate space needed.
IMU (BNO085)	1	28 × 16 × 4 each	mounted in the pelvis (near CoM).
48 V → 5 V buck (5 A)	1	65 × 35 × 18	On PDB, powers Jetson + all ESP32s consider airflow.
48 V → 12 V buck (3 A)	1	55 × 30 × 15	On PDB, powers Jetson fan + auxiliaries.
12S Battery Management System	1	90 × 60 × 18	Mounts adjacent to battery pack, ideally inside the enclosure or clipped to its exterior.
Bulk capacitors (1000 μ F 63 V)	2	20 dia × 42 tall each	On PDB, radial lead cans standing upright.
XT90-S battery connector	1	20 × 33 × 15	Panel mounted on torso exterior for battery swap. Pigtail to PDB.
40 A ANL fuse holder	1	95 × 32 × 30	Inline between battery XT90-S and PDB 48V input.
E-stop switch	1	Ø22 hole, 45 mm behind panel	Panel mounted on upper rear of torso ("shoulder blade" area) — must be reachable while robot is walking cuts 48 V motor rail only.
LiPo cell voltage monitor	1	45 × 22 × 12	Small clip mount near BMS.
microSD card (Jetson boot)	1	in Jetson socket	No separate space, ensure socket is accessible without full disassembly.

2. Battery & Enclosure

Battery pack

- **Chemistry:** 12S LiPo (44.4 V nominal, 50.4 V full charge). 2 pack
- **Capacity target:** ~5000 mAh (222 Wh total). Rough per pack envelope: 155 × 45 × 45 mm each.
- **Total pack envelope (2 packs stacked):** ~155 × 45 × 90 mm.
- **Total pack envelope (2 packs side by side):** ~155 × 90 × 45 mm.

- **Preferred orientation:** as low in the torso as possible (drops CoM, improves locomotion stability).

Enclosure requirements

- **Rigid, load-bearing compartment** integrated into the torso frame. Must survive a full 1.3 m sideways/backward fall onto concrete without deforming.
- **Internal padding:** 3–5 mm closed cell foam all around..
- **Retention:** strap, latch, or bolt down lid pack should not shift during a fall.
- **Ventilation:** a small vent slot on top of the enclosure to let pressure out.
- **Access:** hinged or removable lid so battery can be swapped for charging without disassembling the torso.
- **Enclosure envelope with foam and shell:** add ~15 mm to each dimension of the raw pack (i.e. plan ~185 × 120 × 60 mm for a stacked-pack config).

3. Access & Serviceability

- **Battery swap:** panel or lid
- **E-stop:** reachable while robot is standing/walking
- **Jetson USB / HDMI / RJ45:** accessible without full disassembly (small side door or grommet).

5. V2 Reservations (Head Planned for Fall)

- **Neck cable channel:** reserve a ~15 mm through channel from the top of the torso where the neck will mount, sized for ~5 wires.
- **PDB footprints for second buck + Ethernet header:** electrical will handle this on the PCB, no mechanical impact.

v2 head components (not designed for now)

- **Head Jetson Orin Nano:** same envelope as body Jetson (~100 × 79 × 45 mm).
- **USB microphone array:** ~65–100 mm dia disc, ~15 mm tall.
- **Amplified speaker (3 W):** ~50 × 35 × 25 mm + small amp board.
- **IMX219 CSI camera:** ~25 × 24 × 9 mm on a flex ribbon (front-facing).